

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3 HT	(a)			Correct symbols for lamp and cell and variable resistor shown correctly in a series circuit (1) Accept variable power supply instead of cell and variable resistor If more than 1 cell shown they must be connected correctly in terms of polarity and no line passing through the cell Correct symbol for voltmeter and connected in parallel with their symbol for a lamp only (1) Correct symbol for ammeter and connected in series (1)	3			3		3
	(b)	(i)		Scale on x -axis of 1 V/cm and on y -axis of 0.1 A/cm (1) Accept scale on y -axis of 0.125 A/cm All points correctly plotted $\pm <1$ small square division – ignore (0,0) plot (1) Less than 5 points correctly plotted $\pm <1$ small square division – ignore (0,0) plot (0) Smooth curve between 0 – 12 V with $\pm <1$ small square division (1)		3		3	3	3

Question				Marking details	Marks Available																	
					AO1	AO2	AO3	Total	Maths	Prac												
		(ii)		Any one calculation of resistance e.g. $R = \frac{2}{0.4} = 5 \text{ } [\Omega] \text{ (1)}$ Another calculation of resistance e.g. $R = \frac{12}{1.1} = 10.91 \text{ } [\Omega] \text{ (1)}$ <u>so disagree</u> – conclusion must be consistent with results and based on at least 2 calculated values of resistance (1) For information: <table><tr><th>Voltage (V)</th><th>Resistance (Ω)</th></tr><tr><td>2</td><td>5[.00]</td></tr><tr><td>4</td><td>5.26</td></tr><tr><td>6</td><td>6.38</td></tr><tr><td>10</td><td>9.26</td></tr><tr><td>12</td><td>10.91</td></tr></table> Ignore incorrect rounding of resistance values Use of $R = \frac{P}{I^2}$ to obtain correct values award the marks	Voltage (V)	Resistance (Ω)	2	5[.00]	4	5.26	6	6.38	10	9.26	12	10.91			3	3	3	3
Voltage (V)	Resistance (Ω)																					
2	5[.00]																					
4	5.26																					
6	6.38																					
10	9.26																					
12	10.91																					
				Question 3 total	3	3	3	9	6	9												